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PAPER

Risk-Aware Template-Switching Control under Partial Quantum Observability

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Abstract

With only partial observations, a discrete reference template can fit the available data yet still be a harmful choice. We ask whether a test based on observed data can filter such template switches without access to the target state. An audit of our submitted three-qubit simulations found a degenerate clock-conditioned representation and a veto driven solely by an uncertainty threshold. We retain those results as an audit, without attributing their effect to relational spectral structure. R1 establishes an energy-matched finite-clock construction. R3 compares a continuous estimate with a direct nearest-template estimate and tests score, uncertainty, and held-out residual switching rules. Both candidates are fitted on four clock labels and scored unchanged on the other four. We evaluate one corrected run and three independent replications, each with separate reference-bank, calibration, and test sets. Because any one-sided veto reduces the unconditional number of harmful switches, we additionally compare selection quality at equal expected switching rates. In a reviewer-prompted analysis of the same four streams, retaining half of the score-only proposals by held-out residual lowers the harmful fraction among accepted switches by 20.8–23.7 percentage points relative to uncertainty ranking. The combined gate has a smaller, operating-point-dependent benefit. A continuous estimator optimized using all measurements nevertheless achieves higher grid-mean fidelity than these switching rules. The contribution is a controlled assessment of selection information and measurement cost in finite-data template switching. The clock specifies a physically consistent observation model; neither its necessity for the decision task nor a relational-spectral performance advantage is established.

1 Introduction

Quantum error correction relies on diagnostic information that can be incomplete or noisy. Syndrome-based correction, recovery maps, autonomous protection, and error mitigation address different parts of this problem [1, 2, 3, 4, 5]. Finite-shot state estimation raises a related decision question: when should a policy replace a continuous estimate with a discrete reference template? The template may fit the available counts yet yield lower target fidelity. We test whether counts withheld from fitting can filter these harmful switches. This decision test does not measure the efficacy of a syndrome decoder.

The submitted work proposed to derive a switching signal from clock-conditioned graph spectra. The audit reported here shows why that interpretation fails: the conditional density is constant, and the admissibility condition never binds. We document the submitted construction and test a held-out residual gate using an energy-matched finite clock and finite-shot counts.

Graph and spectral representations remain useful descriptive tools for the submitted audit [6, 7, 8]. The R3 safety test instead asks whether a template fitted on four clock labels predicts the other four as well as a continuous estimate. This finite-data decision is complementary to characterization and uncertainty quantification [9, 10, 11]. It does not evaluate a syndrome decoder or channel-level mitigation method.

R3 evaluates this question in a controlled encoded logical-qubit testbed. Candidate A is a continuous Bloch-ball projected least-squares estimate and candidate B is the direct nearest

template under the same fitting-label loss. A policy may accept or veto a proposed switch from A to B. The held-out gate can only veto a proposal; it cannot create a template switch. The comparison therefore exposes both harmful-switch reduction and any loss of mean fidelity relative to the continuous candidate.

R1 supplies an energy-matched finite-clock construction with nonconstant conditional states. R3 tests the switching decision with physical code noise, direct-template and uncertainty-only controls, and a corrected held-out residual gate. Each of its four fixed streams has separate bank, calibration, and test targets; uncertainty intervals resample target pairs and measurement seeds. The residual gate reduces harmful switches relative to score-only selection in every evaluated cell, but the subset relation alone cannot establish useful selection. R4 therefore compares the same records at equal expected switching rates and adds globally optimized continuous-loss estimators. This distinguishes information about proposal quality from a reduction in intervention frequency.

The contribution is a controlled benchmark of a specific decision: whether to replace a fitting-stage continuous estimate by a fixed-bank template. The held-out residual contains selection information beyond random rejection and uncertainty in this testbed, while the full-measurement continuous control exposes the cost of reserving data for that decision. R1 supplies the physically consistent observation model needed for the adopted clock interpretation; it is not a separate performance contribution. The validation principle is ordinary data splitting, applied here to a specified quantum measurement model with explicit candidate, information, and switching-rate controls.

2 Background and Positioning

2.1 Clock-conditioned relational dynamics

Relational formulations of quantum dynamics describe change through correlations with an internal clock rather than an externally supplied time parameter. This idea appears in background-independence and relational-dynamics discussions [12, 13]. The Page–Wootters mechanism gives the operational ingredient used here: a constrained global state can be conditioned on a clock subsystem to obtain relative system states [14, 15, 16]. Related work has studied robustness, interactions, noncausal circuits, measurement-related time flow, and computational uses of quantum time [17, 18, 19, 20, 21]. Our use of this literature is narrow. R1 uses clock-conditioned slices to construct a finite observation model, and R3 uses its labels to split fitting and validation records. The switching rules operate on the resulting count record. An equivalent randomized-setting experiment has the same likelihood, as shown in Section 4.1; the internal-clock interpretation is not required by the statistical decision rule.

2.2 Spectral graph representations of quantum structure

Graph-based and spectral representations provide compact descriptions of relational structure. In quantum information, graph Laplacians have been related to density matrices, separability, and quantum-state representations [6, 7, 22]. More broadly, quantum network models provide a language for connecting graph structure, correlations, and quantum dynamics [8]. These works motivate the use of spectral signatures as auditable summaries of relational connectivity.

The submitted work differed from graph-state or network modeling approaches in its use of clock-conditioned spectral families. A single graph spectrum was not the central object; each internal clock label produced a relational graph, and the ordered family of Laplacian spectra defined a relational spectral trajectory. The revision retains that construction only to make the failed submitted-grid mechanism auditable. It is not an evidence source for R3.

2.3 Quantum recovery, mitigation, and control

Quantum recovery and error mitigation have developed along several complementary directions. Recovery-map approaches formalize channel-level reconstruction and approximate reversibility [1], while quantum error-correction experiments and autonomous protection schemes demonstrate how encoded or stabilized systems can preserve quantum information under noise [23, 2, 24]. In the NISQ setting, mitigation methods such as probabilistic error cancellation, virtual distillation, detection-based mitigation, and learning-based correction address noisy circuit outputs without requiring full fault tolerance [3, 25, 26, 4, 5, 27].

The submitted framework placed a policy layer above two candidate recovery mechanisms. In those encoded-QEC simulations, syndrome-first recovery was the decoder-facing baseline, while relational admissible recovery supplied an alternative candidate. R3 asks a different, narrower question: it compares a continuous count-record estimate with a direct template under equal information and uses held-out counts to test the proposed switch.

2.4 Partial observability, uncertainty, and benchmarking

Reliable recovery decisions depend on how much diagnostic information is available and how trustworthy that information is. Benchmarking and characterization methods provide tools for quantifying device behavior and validating performance claims [9]. Randomized-measurement techniques and uncertainty-quantification frameworks also emphasize that finite, noisy, or incomplete observations must be interpreted with calibrated uncertainty [10, 11]. Noise modeling, live noise fingerprinting, robust control, and non-Markovian noise studies further show that quantum behavior can depend strongly on the structure and timing of noise processes [28, 29, 30, 31, 32, 33].

The submitted grid tested candidate selection when syndrome records were degraded, using a simple syndrome-stress index. Those diagnostics are retained for the audit. R3 instead tests template switching with finite-shot counts and a held-out residual. Its uncertainty statistic is a thresholded proxy, not a calibrated posterior. Both analyses report harmful switches and beneficial switches that the policy blocks.

3 Submitted-implementation audit

The submitted implementation is retained as an audit, with its full construction, penalties, metrics, and regime tables in Supplementary Sections S1–S10. It used $H_S = 0$ together with a clock initialization and Hamiltonian that produced constant normalized conditional system densities. Its nominal joint state also failed the stated zero-energy constraint. Consequently, those simulations did not instantiate the nondegenerate relational trajectory required by the proposed interpretation.

The submitted admissibility gate was nonbinding in every evaluated row. After removing the uncertainty conjunct, the score/admissibility-only counterfactual vetoed no proposed switches. Historical improvements in downside metrics therefore arose from the uncertainty-threshold veto and cannot be attributed to relational-spectral admissibility. Tables S11–S14 retain the intervention counts, uncertainty intervals, and row-level effects so that this withdrawal is traceable.

The candidate comparison also had asymmetric information: the submitted syndrome-first candidate used the complete recovery channel, while degraded syndrome symbols changed controller support rather than the candidate itself. It was therefore not a damaged-syndrome decoder benchmark. R1 and R3 replace these premises with a valid finite observation model and candidates fitted from identical finite-shot counts. R4 then asks whether the observed gate effect survives an explicit comparison at equal switching rates.

4 Observation model and template-switching assessment

4.1 Physically specified finite-clock construction (R1)

The revision experiments use an encoded logical qubit and an energy-matched two-level clock, rather than the submitted $H_S = 0$ setup. Let

$$H_C = \omega Z_C, \qquad H_S = -\omega(n_x X_L + n_z Z_L), \qquad \omega = 0.8, \tag{1}$$

where the tilted setting is $n = (1, 0, 1)/\sqrt{2}$. If $|n_+\rangle$ and $|n_-\rangle$ are the logical eigenstates of $n_x X_L + n_z Z_L$ and $|\psi\rangle = c_+|n_+\rangle + c_-|n_-\rangle$, we prepare

$$|\Psi\rangle = c_+|0\rangle_C|n_+\rangle_L + c_-|1\rangle_C|n_-\rangle_L, \tag{2}$$

which obeys $(H_C + H_S)|\Psi\rangle = 0$. Eight clock effects $E_k = (2/8)|\tau_k\rangle\langle\tau_k|$ at $\tau_k = k\pi/(8\omega)$ form the finite POVM. R1 verifies the null constraint, conditional evolution, POVM completeness, nondegenerate clock geometry, and the rank-three tilted observation design. It is an observability qualification, not a QEC efficacy result.

The physical role of this construction must be separated from the statistical role of the gate. For the population-count readout used here, the likelihood of clock label k and outcome $z \in \{-1, +1\}$ can be written

$$p(k, z \mid \boldsymbol{r}) = \frac{1}{16}\left[1 + z(1 - 2\eta)\boldsymbol{d}_k^{\mathsf{T}}\boldsymbol{r}\right], \tag{3}$$

where $\boldsymbol{r}$ is the initial logical Bloch vector, η is the known readout-flip probability, and $\boldsymbol{d}_k$ includes the declared rotation and physical noise. Selecting one of eight ordinary measurement settings uniformly and measuring its corresponding logical population yields the same likelihood. Thus energy matching is required for our chosen stationary-state clock interpretation, but no clock-specific statistical advantage or operational necessity follows. Graph spectra do not enter R3 or R4.

4.2 Corrected held-out template-switching risk test (R3)

R3 uses four fixed data streams, each with disjoint reference-bank, calibration, and held-out target sets. Stream 0 reuses R2's random streams to isolate the gate-definition correction; streams 1–3 use independent bank, calibration, target, and measurement streams. The switching policy receives finite-shot count records, the known dynamics, and a fixed 32-state bank. Four clock labels are used for fitting, and the other four for validation. The bank, calibration, and test targets are pure logical-qubit states. Dynamics, physical noise, and readout-flip probabilities are known to the estimators. Candidate A is a continuous Bloch-ball projected weighted least-squares estimate; candidate B is the direct nearest template under the same fitting-label loss.

Let $L_{\mathrm{val}}(X)$ be the weighted squared loss of the unchanged fitting-label candidate X on the validation-label counts. The held-out residual statistic is

$$h = L_{\mathrm{val}}(B_{\mathrm{fit}}) - L_{\mathrm{val}}(A_{\mathrm{fit}}). \tag{4}$$

Neither candidate is refit on validation labels. Score-only C_2, uncertainty-only U, held-out-residual-only H, and the combined gate C_3^R are evaluated on identical records. For each stream, the score, uncertainty, and residual thresholds are set once to the 75th empirical quantiles of that stream's separate calibration set at the fixed bit-flip, 1,024-copy, physical-noise 0.05, readout-flip 0.05 operating point. They remain fixed across all 24 held-out test cells in that stream; their values may differ between streams. The target density is used only after a decision to calculate fidelity. An oracle nearest-template fidelity is reported only as an offline bank-resolution diagnostic, never as a policy input.

In every one of the 24 grid cells in each of the four streams, $H - C_2$ has a negative harmful-switch-rate difference whose two-way target-pair/measurement- seed 95% interval excludes zero. At the fixed middle operating point (bit-flip, 1,024 prepared copies, physical-noise probability 0.05, and readout flip 0.05), the $H - C_2$ differences are -14.844, -8.333, -10.677, and -5.990 percentage points in streams 0–3. The two-way interval for $C_3^R - U$ is negative in 12, 8, 12, and 6 cells, respectively; the combined gate therefore does not improve uniformly on uncertainty-only selection. The sign of the $C_3^R - A$ mean-fidelity difference also varies across streams. Full results and intervals are reported in the Supplementary Information.

These unconditional harmful-switch reductions are descriptive: H accepts a subset of the C_2 switches, and C_3^R a subset of the U switches. They cannot alone establish that the rejected proposals are selectively worse. R4 addresses this issue directly.

4.3 Selection quality at equal switching rates (R4)

R4 is a reviewer-prompted secondary analysis of the existing R3 streams. Its analysis rules were fixed before computing the new comparisons, after the original R3 outcomes were known. It replays all 73,728 target/count records from their recorded seeds and verifies agreement with the saved R3 diagnostics and fidelities; no new count realizations are introduced. Within each cell, we first thin the original C_2, H, and U decisions uniformly to the expected number accepted by C_3^R. Averaging over this randomization gives identical expected switching rates. Only the twelve 1,024-copy cells per stream have positive common coverage; at 256 copies, the original U and C_3^R accept no switches. The harmful fraction among accepted switches, rather than its unconditional count, now measures selection quality. Paired two-way bootstrap samples recompute the thinning factors.

At common coverage, the stream-mean harmful fraction for H is lower than for uniformly thinned C_2 by 7.62, 3.00, 6.19, and 4.48 percentage points in streams 0–3. The corresponding $C_3^R - U$ reductions are 5.23, 1.46, 4.37, and 1.93 points. All eight paired pointwise 95% intervals exclude zero (Supplementary Table S17). These contrasts establish selection information in the evaluated positive-coverage cells, beyond the automatic effect of accepting fewer switches.

We also compare batch rankings that accept 25%, 50%, or 75% of the original C_2 proposals. Scores are fitting loss difference, uncertainty u, residual h, and the combined statistic $\max(u/t_u, h/t_h)$ using the original R3 calibration thresholds. Uniform random retention supplies a control. Boundary ties are randomized analytically, so expected switching rates match exactly in each cell and resample. Cutoffs use the batch's observed diagnostics without target outcomes; these are batch selection curves, not newly calibrated deployment rules. Figure 1 shows that residual ranking is more selective than uncertainty or their conjunction over all three retention levels in the stream averages. At 50% retention, its $H - U$ harmful-fraction differences are -20.78, -21.76, -23.68, and -23.50 percentage points, with all four paired intervals below zero.

The numerical harm criterion $F_B < F_A - 10^{-12}$ counts even very small losses. A follow-up sensitivity analysis keeps every decision rule fixed and applies absolute fidelity margins 0.001, 0.005, and 0.01. At margin 0.01 and 50% retention, the $H - U$ harmful-fraction differences are

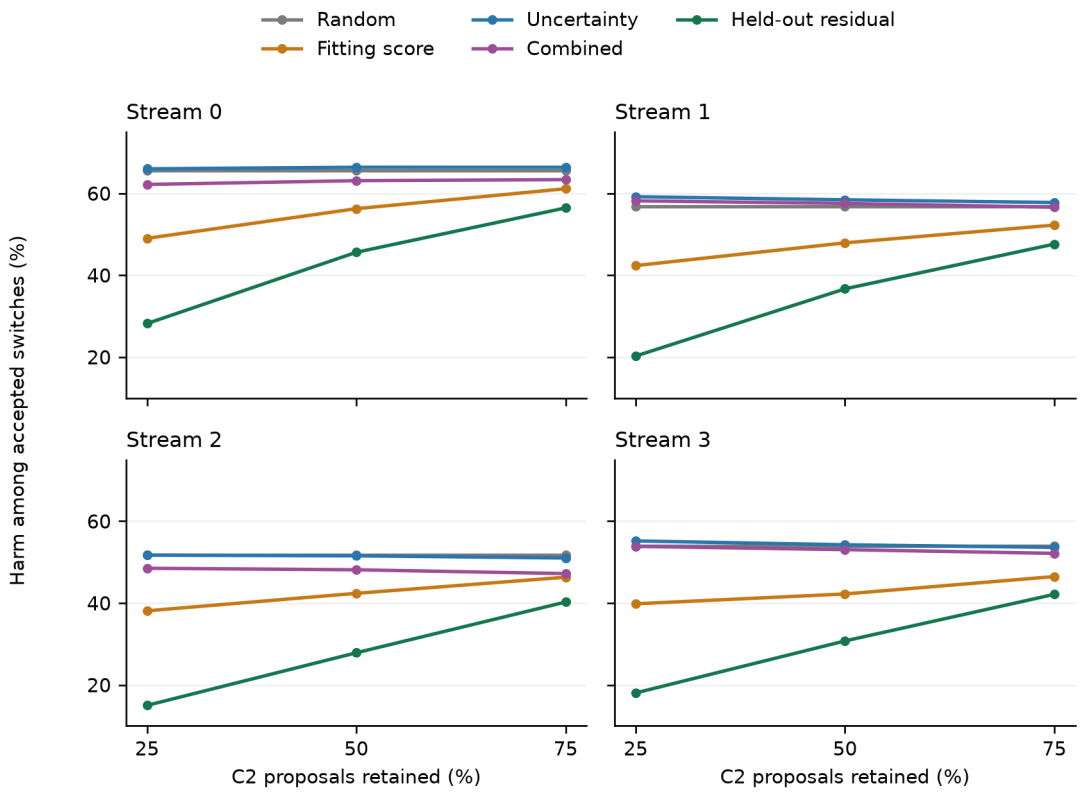


Figure 1. R4 selection quality at identical expected switching rates. Each point equally averages the harmful fraction among accepted switches over all 24 cells of one stream. Retention is relative to the original C_2 proposal set, so every rule has the same expected number of switches within a cell. Lines show point estimates; paired intervals and the distinction between original fixed gates and batch rankings are given in Supplementary Tables S17–S18. The plotted harm criterion is $F_B < F_A - 10^{-12}$; margin sensitivity and direct mean-fidelity comparisons are in Tables S20–S21. These are secondary analyses of the R3 records.

-26.60, -24.08, -23.80, and -25.24 percentage points in streams 0–3, with all four paired intervals below zero. Table S20 reports every declared margin. These margins assess sensitivity; they are not validated application-specific utility thresholds.

The direct $H - U$ overall mean-fidelity differences at 50% retention are 0.01084, 0.00977, 0.00748, and 0.01015, with all four paired intervals above zero (Table S21). These signed-fidelity comparisons retain all gains and losses, irrespective of the harm margin. They show value within the specified batch-selection task, without implying the same gain for a calibration-only deployment policy.

4.4 Continuous estimation and the scope of template use

The original A radially projects an unconstrained weighted least-squares fit onto the Bloch ball. That projection need not minimize the constrained loss. R4 therefore adds its global constrained minimizer A^*_{fit} and a full-record control A^*_{all}, both solving

$$\min_{\|\boldsymbol{x}\|\le 1} \sum_{k\in\mathcal{I}} n_k \left(\boldsymbol{d}_k^{\mathsf{T}}\boldsymbol{x} - y_k\right)^2, \tag{5}$$

with $\mathcal{I}$ equal to the fitting labels or all eight labels. The latter uses the same total prepared-copy budget as the residual gate and accounts for the opportunity cost of holding measurements aside. Feasibility, optimality conditions, and loss improvement over A are checked for every fit. Optimization concerns this specified count loss, not the unknown target fidelity.

At 50% retention, the H ranking has higher grid-mean fidelity than A^*_{fit} in each stream, with differences 0.00522, 0.00815, 0.00676, and 0.00847 and paired intervals above zero. However, its differences from A^*_{all} are -0.01195, -0.00897, -0.01000, and -0.00949, with intervals below zero (Supplementary Table S19). Thus useful selection within a fitting-stage template task does not establish an advantage over continuous estimation using all available measurements.

A discrete template is an optional, restricted output in this study. A finite action alphabet could motivate such a decision in an application, but we have not measured a hardware cost or

downstream utility that would justify that restriction. When an unrestricted continuous estimate and target fidelity are the only objectives, the full-record continuous control is the supported choice on this grid. The results characterize when residual evidence helps select template proposals; they do not recommend template substitution as a general recovery procedure.

5 Conclusion

The audit found that the submitted clock construction was degenerate and that its reported veto effect could not be attributed to admissibility. We retain those results as an audit. The revision adds an energy-matched finite-clock construction and a template-switching experiment with separate calibration and held-out target sets.

The supported result concerns selection quality in a specified two-candidate task. After matching expected switching rates, held-out residual evidence selects fewer harmful template substitutions than random retention or uncertainty alone in the reported stream averages. This adds evidence beyond the unconditional harmful-switch reduction guaranteed by a one-sided veto.

The extra complexity of the combined gate is therefore not generically earned relative to uncertainty-only selection. Across streams 0–3, its unconditional harmful-switch-rate interval is strictly below that of the uncertainty-only gate in only 12/24, 8/24, 12/24, and 6/24 evaluated cells, respectively, and its mean-fidelity difference from the continuous estimate changes sign between streams. R4 finds a smaller but measurable conditional-quality benefit over U at matched coverage in the active 1,024-copy cells; nevertheless, residual-only batch ranking is better than the combined ranking in the stream averages. These comparisons do not justify preferring the extra uncertainty conjunct within the tested scaling family.

All-measurement continuous fitting gives higher grid-mean fidelity than the tested switching rankings, so the study provides no general reason to prefer a discrete template. The finite clock supplies a consistent physical realization of the count model, with no demonstrated necessity for the selection rule. The uncertainty statistic remains an uncalibrated proxy. R4 reuses the four R3 streams of pure logical-qubit targets and its intervals are pointwise and conditional on the fixed banks, known observation model, and thresholds. Mixed targets, model misspecification, out-of-grid generalization, an application-specific reason for discrete outputs, and channel-level recovery require separate evidence.

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Competing interests

The authors declare no competing interests.

Data availability

The R3 source code, protocol used for the runs, raw outputs, and submitted-grid audit artifacts are available at https://github.com/Jeong-Jaeyun/Risk-Aware-Relational-Spectral-Recovery-under-Partial-Quantum-Observability_Phase_1/tree/submission/revision-r3-clean. The R4 analysis protocol, source, input records, and complete cell summaries are supplied in the accompanying reproducibility archive.

Author contributions

J.Y.J. conceived the study, implemented the simulations, performed the data analysis, and drafted the manuscript. H.Y.J. supervised the research, provided critical intellectual input, and revised the manuscript. All authors have read and approved the final version of the manuscript.

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